

National Tsing Hua University
Department of Electrical Engineering
EE429200 IC Design Laboratory, Fall 2025

**Lab 13: APR Flow with Innovus –
CTS 、Routing 、Verification 、Power Analysis (1%)**

Assigned on Nov. 27, 2025

Due day in **Dec. 4, 2025**

Objective

In this lab, you will learn:

1. How to use Innovus to perform a basic APR flow
2. Post-layout simulation
3. Power analysis with PrimeTime

Demo checklist:

- ☐ Clean DRC, connectivity, and antenna verification
- ☐ Post-layout simulation result
- ☐ Power report*3 by PrimeTime
- ☐ LEC log file by Conformal LEC
- ☐ Questions:
 1. What are functions of core fillers?
 2. Please refer to *lec/post_layout_lec/lec.log*, why are there 35 unmapped DFFs in Golden (i.e. gate-level netlist)? Does it mean LEC comparisons fail?
 3. What are purposes of three PrimeTime power reports? (Hint: a) there are many parameters during powerplan, such as # of core rings and # of stripes. b) waveform of post-layout simulation is difficult to obtain in some designs.)

Action Items

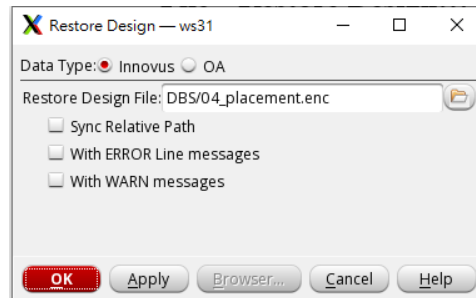
I. Clock Tree Synthesis

1. In the *lab11/innovus/run*, open the GUI of Innovus by:

\$ innovus

- Open design from the previous step.

“File->Restore Design...”



- Remove clock latency from design constraint

```
$ innovus > set_interactive_constraint_modes [all_constraint_modes]
$ innovus > reset_clock_latency [all_clocks]
```

- Run CTS (it might take about 20 minutes)

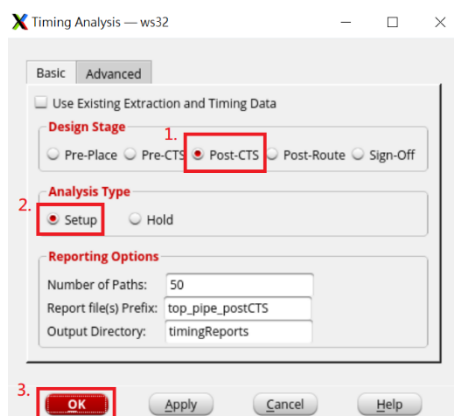
Understand the command in *innovus/pre_layout/ccopt.cmd*

```
$innovus > source ../pre_layout/ccopt.cmd
```

- Report timing

(a.) Setup time

“Timing->Report Timing...”

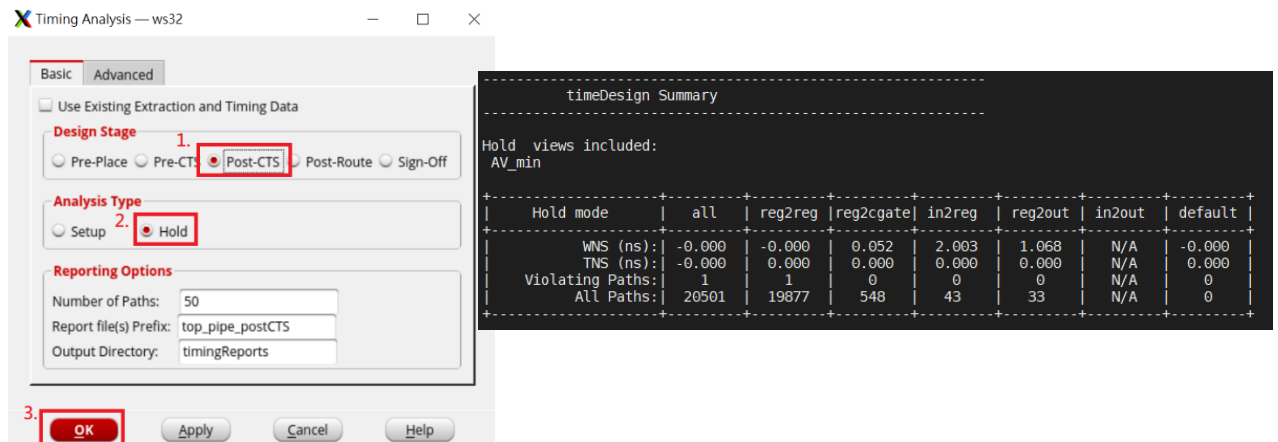


timeDesign Summary							
Setup views included: AV_max							
Setup mode	all	reg2reg	reg2cgate	in2reg	reg2out	in2out	default
WNS (ns):	2.031	2.031	4.392	7.882	8.531	N/A	0.000
TNS (ns):	0.000	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	0	N/A	0
All Paths:	20501	19877	548	43	33	N/A	0
DRVs	Real		Total				
	Nr nets (terms)	Worst Vio	Nr nets (terms)				
max_cap	0 (0)	0.000	0 (0)				
max_tran	28 (56)	-0.723	28 (56)				
max_fanout	600 (600)	-45	816 (816)				
max_length	0 (0)	0	0 (0)				

Check if setup time meets the timing requirement.

(b.) Hold time

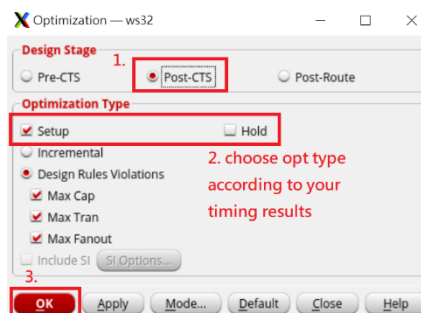
“Timing->Report Timing...”



Check if hold time meets the timing requirement.

6. (Optional) Optimize Design (if the timing did not pass during above steps.)

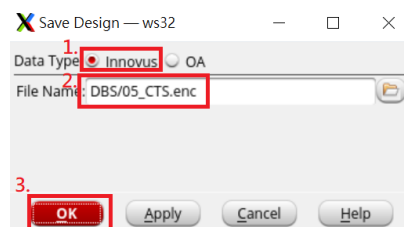
“EOC->Optimize Design...”



Then repeat 6-(a) or 6-(b) to check your design meets the timing requirements.

7. Save Design

“File->Save Design...”



II. Route

1. Run Routing (it might take about 20 minutes)

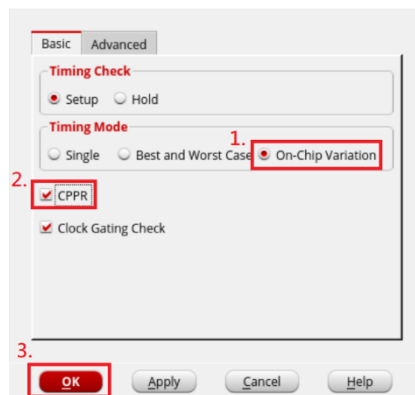
“Route->NanoRoute->Route...”



2. Check Timing

(a.) Firstly, we need to change the timing analysis mode.

“Tool->Set Mode->Specify Analysis Mode...”



(b.) Activate SI aware

`$innovus > setDelayCalMode -SIAware true`

5. Verification

Check that no violation is reported in the terminal

```
VERIFY DRC ..... Sub-Area: {432.000 427.200 513.600 512.240} 36 of 36  
VERIFY DRC ..... Sub-Area : 36 complete 0 Viols.  
  
Verification Complete : 0 Viols.  
  
*** End Verify DRC (CPU: 0:00:31.0 ELAPSED TIME: 33.00 MEM: 86.1M) ***
```

(1) DRC

\$ innovus > verify_drc

(2) Connectivity

“Verify->Verify Connectivity”, cancel DanglingWire (Antenna),

OK

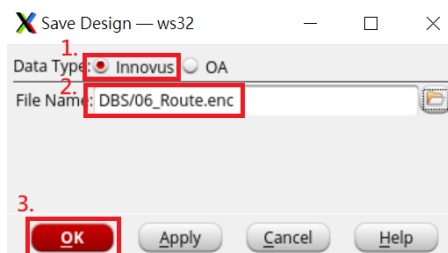
(3) Antenna

“Verify->Verify Process Antenna”,

OK

6. Save Design

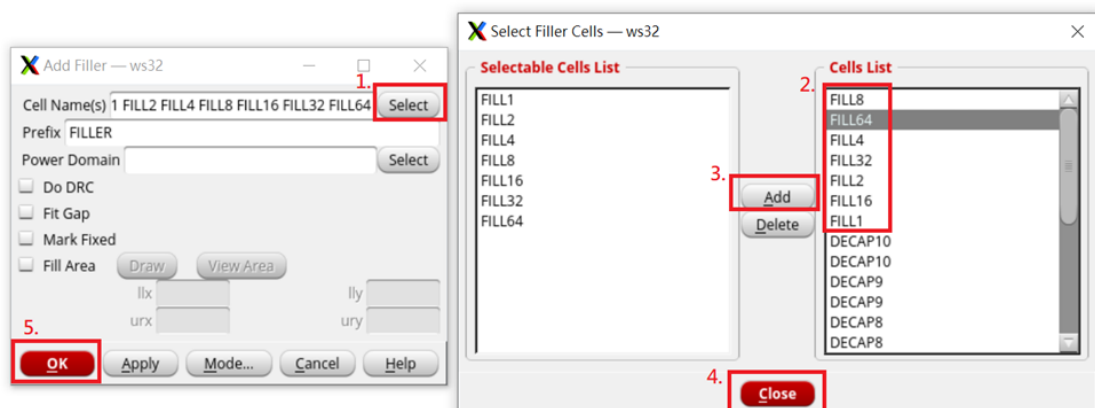
“File->Save Design...”



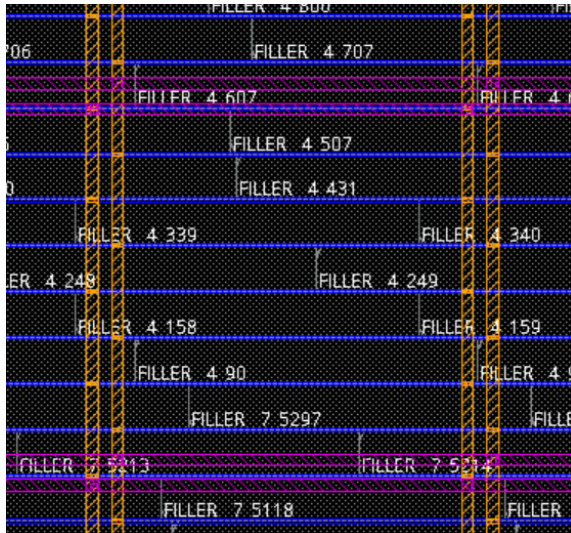
III. Design for Manufacturability (DFM)

1. Add Filler

“Place->Physical Cells->Add Filler...”



Check if regions without standard cells in your layout have been filled by the fillers.



IV. Export design

1. Export the netlist and the corresponding timing data in SDF (Standard Delay Format) file format for post-layout simulation

```
$ innovus > source ../../ref_script/savesdf.cmd
```

2. Export GDSII layout

```
$ innovus > source ../../ref_script/savegds.cmd
```

Note: This is the layout which will be processed and verified by other tools in a normal backend flow. But these operations won't be involved in the following lab.

3. Export standard design constraint (SDC) and parasitic information (SPEF) for power simulation

```
$innovus > write_sdc ../post_layout/CHIP_layout.sdc
```

```
$innovus > setExtractRCMode -engine postRoute
```

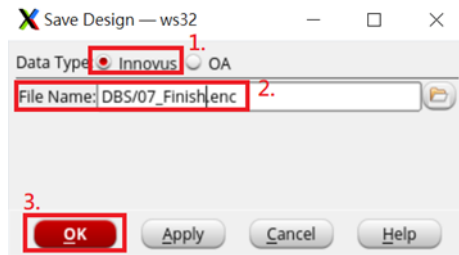
```
$innovus > reset_parasitics
```

```
$innovus > extractRC
```

```
$innovus > rcOut -rc_corner RC -spef ../post_layout/CHIP_layout.gz
```

4. Save Design

“File->Save Design...



For more detailed information about Innovus, type the following command in the terminal

```
$ unix > /usr/cadence/INNOVUS/cur/bin/cdnshelp &
```

V. Post-layout simulation

1. In the *lab11/post_sim/* directory, read these files (postsim.f, test_top.v) and understand how the simulation environment is configured.

Note: In the testbench, we need to annotate the SDF file to the CHIP instance (line 81-83). The SDF file, which is exported by Innovus in the previous step, records the delay information of each gate and net.

2. Run the simulation by execute the following command, and then check whether the simulation result is correct or not

```
$ vcs -R -full64 +v2k -f postsim.f -debug_access+all
```

VI. Logic Equivalence Check

Assume we have verified the logic equivalence between RTL and gate-level netlist before P&R (you can also run LEC by yourself for RTL and gate-level netlist for a completed verification flow). After physical design, we need to check the logic equivalence between gate-level netlist and layout by Conformal LEC since we cannot run all possible cases in post-layout simulation.

1. In the *lab11/lec/post_layout_lec* directory, read all the files and understand how the environment is configured.
2. Run the LEC between gate-level and post-layout netlist by execute the following command, and then check whether the result is equivalent or not.

```
$ sh run_lec.sh
```


VII. Time-based Power Analysis with PrimeTime

Unlike the static power analysis in Design Compiler and Innovus, time-based power analysis estimates the power by the actual waveform. The actual signal activities make the estimation more realistic.

To run the power analysis, you need to prepare the following files:

- netlist(*.v, generated by DC/Innovus)
- Design Constrain file(*.sdc)
- Waveform (*.fsdb), waveform by pre/post-layout simulation
- SPEF file (Standard Parasitic Exchange format, generated by Innovus)

1. Pre-layout power analysis

Generally, this should be done before P&R to determine the power needed. Here we perform pre-layout power analysis after P&R for convenience.

(1) Run gate-level simulation

In **lab11/sim** directory, view gatesim.f and run

```
$ vcs -R -full64 +v2k -f gatesim.f -debug_access+all
```

(2) Run pre-layout power estimation

In **lab11/primetime/pre_layout** directory, view pt_px.tcl and run

```
$ pt_shell -f pt_px.tcl
```

Note: SPEF is not needed here because we don't have physical information about wiring. The SPEF file, which provides important parasitic information to PrimeTime, records parasitic information of the layout netlist.

2. Post-layout power analysis with post-sim waveform

(1) Modify SDC generated by Innovus

Some statements are not allowed in PrimeTime

Open **lab11/innovus/post_layout/CHIP_layout.sdc**

Comment the following lines out

```
# current_design top_pipe
```

```
# set_max_fanout 20 [get_designs {top_pipe}]
```

(2) Run post-layout power estimation with post-sim waveform

In **lab11/primetime/postsim_waveform** directory, **modify** pt_px.tcl and run

```
$ pt_shell -f pt_px.tcl
```

3. Post-layout power analysis with pre-sim waveform

(1) Use the same SDC as the previous step

Make sure SDC file has been properly modified

(2) Generate pre-simulation waveform. In **lab11/sim** directory

```
$ vcs -R -full64 +v2k -f presim.f -debug_access+all
```

(3) Run post-layout power estimation with pre-sim waveform

In *lab11/primetime/presim_waveform* directory, **modify** `pt_px.tcl` and run
`$ pt_shell -f pt_px.tcl`

Note: When we want to use pre-sim waveform for power analysis in PrimeTime, we have to add the argument “-rtl” in “read_fsdb” command (line 20 in `pt_px.tcl`) to specify the VCD/FSDB file comes from RTL simulation.

4. Power estimation Comparison

Compare the power report of the three results above.

	Pre Layout	Post Layout (pre-sim waveform)	Post Layout (post-sim waveform)
Net Switching Power			
Cell Internal Power			
Total Power			

Congratulations!! You have finished all the labs in this semester!!

Appendix

Here is the file list of the lab package. Please check if anything missing!

Directory	Filename	Description
innovus/pre_layout/	top_pipe_syn.v	Synthesized netlist
	top_pipe_syn.sdc	Modified SDC for Innovus
	ccopt.cmd	CTS commands
	mmmc.view	MMMC setting file
sim	instruction.txt	CPU simulation pattern
	run_presim.sh	Shell script for pre-simulation
	presim.f	File list for pre-simulation
	run_gatesim.sh	Shell script for gate-level simulation
	gatesim.f	File list for gate-level simulation
	test_top.v	Testbench for the CPU
hdl	top.v	Top module for this CPU
	top_pipe.v	Pipelined design for this CPU
	ID_stage.v	RTL for ID stage
	IF_stage.v	RTL for IF stage

	controller.v	RTL for control signal
	regfile.v	RTL for register file
	ID_EXE.v	RTL for ID to EXE flip flop
	EXE_stage.v	RTL for EXE stage
	alu.v	RTL for alu module
	CPU_define.v	Definition file
	dsram.v	RTL for memory
	PC.v	RTL for program counter
	IF_ID.v	RTL for IF to ID flip flop
syn	.synopsys_dc.setup	DC setup file
	*.tcl	Synthesis scripts
	*.log	Log files
	run_dc.sh	Shell script for synthesis
syn/report	*.out	Report files
	top_pipe_syn.ddc	Netlist & constraints can be read by DC
	top_pipe_syn.saif	Switching activity interchange format (for power)
	top_pipe_syn.sdc	Synopsys design constraints format
	top_pipe_syn.sdf	Standard delay format
	top_pipe_syn.v	Netlist in Verilog
primetime/pre_layout	pt_px.tcl	Script for pre-layout power analysis
primetime/postsim_waveform	pt_px.tcl	Script for post-layout power analysis with post-sim waveform
primetime/presim_waveform	pt_px.tcl	Script for post-layout power analysis with pre-sim waveform
lec/pre_layout_lec	run_lec.sh	Shell script for LEC
	*.do	Scripts for RTL-to-netlist LEC
lec/post_layout_lec	run_lec.sh	Shell script for LEC
	*.do	Scripts for netlist-to-post-layout LEC